

Codeforces Roadmap

colaman · Rating 1106 → Target 1400+

Generated Jun 2026 · 150 curated problems across 5 topics

Phase 1	Two Pointers, Binary Search	60	High — do this week
Phase 2	DP, Graphs, Data Structures	90	Critical — main gap
Phase 3	Divide & Conquer	30	After Phase 2

■ 1000–1200 Beginner	■ 1300–1400 Intermediate	■ 1500–1600 Advanced	■ 1700–1800 Expert
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PHASE 1 — Quick Wins

Topic: Two Pointers

Why: Mechanical technique learnable in 2 days. Unlocks C-level problems immediately.

Tip: Pattern: maintain left & right pointers, move based on sum/condition. Master this before binary search.

1	Segment Folding	580C	1000	■
2	George and Job	467C	1000	■
3	Nearly Lucky Number	121C	1000	■
4	Mahmoud and Ehab and the bipartiteness	862B	1100	■
5	Karen and Coffee	816D	1100	■
6	Array and segments	1108E	1100	■
7	Two Subsequences	1325B	1100	■
8	Maximum Sum of Digits	1202C	1200	■
9	Company Merger	1017E	1200	■
10	K-th Not Divisible by n	1352C	1200	■
11	Swap and Delete	1469C	1200	■
12	Party	1579C	1200	■
13	Good String	1608C	1200	■
14	Coprime	1742C	1300	■
15	Count Subarrays	1873E	1300	■
16	Sum of Cubes	1490C	1300	■
17	Two Pointers	1791C	1300	■
18	Suffix Array	1063F	1300	■
19	Ciel the Commander	321C	1300	■
20	Skyscrapers	1041D	1400	■
21	Minimum Domino Rotations	1033C	1400	■
22	Three Parts of the Array	1197C	1400	■
23	Sum of Medians	1005E	1400	■
24	Sherlock and Array	768B	1400	■
25	Ehab and a 2-op task	1364B	1500	■
26	Remove Duplicates	1073C	1500	■
27	Codeforces Round Pleasures	1060C	1500	■
28	Maximise The Array	1405C	1500	■

29	String Reversal	1416C	1600	■
30	Kth Largest Value	1535C	1600	■

Topic: Binary Search

Why: 6 problems solved is dangerously low. Binary search on answer unlocks 1300-1600 problems.

Tip: Key pattern: 'binary search the answer' — define a `check(mid)` function, find smallest/largest valid `mid`.

1	Vasya and String	676C	1000	■
2	Interesting Drink	507B	1000	■
3	Sagheer and Nubian Market	812C	1100	■
4	Kth Largest Value	1535C	1100	■
5	Points and Powers of Two	1033B	1100	■
6	Present	1322C	1200	■
7	Square Subsets	895C	1200	■
8	Solve The Maze	1356C	1200	■
9	Snacktower	1208C	1200	■
10	Interesting Function	1602C	1200	■
11	Search in Logarithmic Time	1525C	1300	■
12	Count Triangles	1355C	1300	■
13	Construct The String	1543B	1300	■
14	Boolean Function	1638C	1300	■
15	Nezzar and Binary String	1659D	1300	■
16	Beautiful Matrix	364C	1400	■
17	Minimal Segment Cover	1367C	1400	■
18	Sorting by Subsequences	1061C	1400	■
19	Guess The Maximums	1363C	1400	■
20	Maximise The Array	1405C	1400	■
21	Increasing by Modulo	1407C	1500	■
22	Dima and a Bad XOR	1244C	1500	■
23	Decreasing Debts	1238D	1500	■
24	Card Deck	1187D	1500	■
25	Codeforces Round 659	1384C	1500	■
26	Orac and LCM	1737C	1500	■
27	Nastya and Strange Function	1583C	1600	■
28	XOR and Favorite Number	617E	1600	■

29	Counting Points	570E	1700	■
30	Robot on the Board 2	1033F	1800	■

Topic: Dynamic Programming

Why: Only 10 DP problems out of 253 (4%). DP appears in every Div.3/4 from problem C onwards.

Tip: Learn in order: 1D DP → prefix DP → knapsack → 2D DP → DP on strings. Don't skip steps.

1	Covered Points Count	1201C	1000	■
2	Tanya and Postcard	1359C	1000	■
3	Nastia and a Good Sequence	1701C	1000	■
4	Polycarp and Sums of Subsequences	1468B	1100	■
5	Filling Shapes	1209C	1100	■
6	Boredom	455C	1100	■
7	Woodcutters	545C	1100	■
8	GCD Table	582C	1200	■
9	Vasya and Multisets	1051D	1200	■
10	Keshi Is Throwing a Party	1734C	1200	■
11	Team Building	1316C	1200	■
12	Treasure Hunting	1519C	1200	■
13	Vladik and Cafes	811C	1300	■
14	Nested Segments	652D	1300	■
15	Cyclic Components	977E	1300	■
16	Maximum White Subtree	1324F	1300	■
17	Vasya and a Tree	1076E	1400	■
18	Sum of Digits	862C	1400	■
19	Making It Increasing	1437C	1400	■
20	Ants	704B	1400	■
21	Ehab and a Special Coloring	1368C	1400	■
22	Connected Graph	1167D	1500	■
23	Move and Swap	1463E	1500	■
24	Tokitsukaze and Stalagmite	1628D	1500	■
25	Ants Colony	704D	1500	■
26	Palindrome Partition	906E	1600	■
27	Book of Evil	337D	1600	■
28	Rearrange the Array	1374D	1600	■

29	Strange Beauty	1555C	1700	■
30	Minimize the Permutation	1374E	1800	■

Topic: Graphs & BFS/DFS

Why: Only 1 graph + 1 DFS problem. Completely unavoidable above 1300.

Tip: Order: BFS (shortest path) → DFS (connected components) → flood fill → cycle detection → then trees.

1	Journey to the Center of Earth	196C	1000	■
2	Checkposts	380C	1000	■
3	Longest Path in a Tree	1654C	1000	■
4	Fox and Name	388C	1100	■
5	Xenia and Tree	342E	1100	■
6	Berland Trip	1271C	1100	■
7	BFS on Tree	1555B	1100	■
8	Queue in the Train	713D	1200	■
9	Kevin and Permutation	1547D	1200	■
10	Rescue the Princes	1606C	1200	■
11	Xenia and Colorful Gems	342C	1200	■
12	Mashmokh and Reverse Operation	414D	1200	■
13	Bear and Prime 100	660D	1300	■
14	Codeforces Graph Contest	1000B	1300	■
15	Strange Beauty	1555C	1300	■
16	Ehab and the Expected XOR Problem	1364D	1300	■
17	Kevin and Arithmetic	1547C	1300	■
18	Restore the Permutation	1176C	1400	■
19	AND Segments	1101G	1400	■
20	Graph Coloring	1066D	1400	■
21	Greg and Graph	295E	1400	■
22	Anton and Tree	734E	1500	■
23	Tourist	340E	1500	■
24	Roads not only in Berland	196E	1500	■
25	Longest Increasing Subsequence	269B	1500	■
26	Codeforces 1000 F	1000F	1600	■
27	Ehab and the Parity	1364E	1600	■
28	Arpa and a game with Mojtaba	850C	1600	■

29	Restore the Queue	1176E	1700	■
30	Codeforces 1027 G	1027G	1800	■

Topic: Data Structures

Why: Only 4 problems. Prefix sums alone unlock 10+ problems you're currently skipping.

Tip: Order: prefix sums → difference arrays → stack/queue tricks → then segment tree/Fenwick.

1	Ehab and Another Construction	1364A	1000	■
2	Little Elephant and Array	220C	1000	■
3	Sereja and Brackets	380E	1000	■
4	Sonya and Queries	714D	1100	■
5	Colorful Bricks	1027B	1100	■
6	Mashmokh and Numbers	415D	1100	■
7	Karen and Coffee	816D	1200	■
8	Vladik and Complicated Contest	811D	1200	■
9	Nastya and a Colorful Broom	1777C	1200	■
10	Interesting Drink	507B	1200	■
11	Queries on a Permutation With Key	1217D	1300	■
12	Keshi Is Throwing a Party	1734C	1300	■
13	Nested Segments	652D	1300	■
14	Ciel the Commander	321C	1300	■
15	Mahmoud and Ehab and the MEX	862E	1400	■
16	Range Minimum Query	1000C	1400	■
17	Ehab and Prefix MEXs	1364F	1400	■
18	Bear and Reversed Rooksort	660F	1400	■
19	Queue on Segments	1108D	1500	■
20	Kevin and Permutation	1547D	1500	■
21	Arpa and a SCC	850D	1500	■
22	Sonya and Problem Without a Legend	714E	1500	■
23	Restore the Permutation by Suffix Array	1063F	1600	■
24	Mahmoud and Ehab and the Xor	862G	1600	■
25	Tree Queries	1702G	1600	■
26	Arpa's weak amphitheater	850E	1600	■
27	Suffix Array	1063F	1700	■
28	HLD	375D	1700	■

29	Sereja and Crows	380D	1800	■
30	Codeforces 652 F	652F	1800	■

Topic: Divide and Conquer

Why: Appears in harder problems once you're past 1400. Builds on binary search skills.

Tip: Classic patterns: merge sort variant, closest pair, D&C; DP. Always think 'split in half'.

1	Ehab and Path-Minimization	1364C	1000	■
2	Make it Palindrome	1256B	1000	■
3	Bear and Prime 100	660D	1100	■
4	Nastya and a Clever Scheme	1701D	1100	■
5	Vladik and Special Sum	811E	1200	■
6	Mahmoud and Ehab and the Xor	862G	1200	■
7	Tree Queries	1324G	1200	■
8	Arpa and a game with Mojtaba	850C	1300	■
9	Array Stabilization	1095D	1300	■
10	Ehab's Last Theorem	1364G	1300	■
11	Anton and Lines	734C	1300	■
12	Restore the Permutation	1176C	1300	■
13	Sequence Transformation	1188C	1400	■
14	Greg and Graph	295E	1400	■
15	Sereja and Squares	380E	1400	■
16	Strange Beauty	1555C	1400	■
17	Tourist	340E	1500	■
18	Codeforces Round 706	1494D	1500	■
19	Largest Divisible Subset	1005D	1500	■
20	Graph Connectivity 1	1000E	1500	■
21	Codeforces 1060 E	1060E	1600	■
22	Move and Swap	1463E	1600	■
23	Palindrome Partition	906E	1600	■
24	Bear and Reversed Rooksort	660F	1600	■
25	Ehab and the Expected XOR Problem	1364D	1600	■
26	Restore the Queue by Merging	1176F	1700	■
27	Sorting	269E	1700	■
28	Sereja and Suffixes	380F	1700	■

29	Codeforces 1535 F	1535F	1800	■
30	Beautiful Arrangement	939E	1800	■

The Plan in One Line

Solve Two Pointers + Binary Search this week, grind DP + Graphs for 4 weeks, then participate in every Div. 3/4 contest and upsolve the hardest problem you miss.

Target	Specialist (1400) within 8–10 weeks
Daily minimum	2 problems/day on weak topics
Contest rule	Every Div. 3 or Div. 4, no skipping
After each contest	Upsolve first unsolved problem within 24h
Your edge	Math + Greedy + Implementation — use these for fast A/B solves
Upsolve queue	44 attempted problems — do 3 per week